

# XING HAN

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## RESEARCH INTERESTS

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I build multimodal foundation models and agents that acquire the core competencies of human experts: integrating heterogeneous evidence as it unfolds over time, reasoning about “what-if” trajectories, and staying reliable as the situation evolves. My work bridges core machine learning methodology with high-stakes applications such as medicine.

- **Multimodal Foundation Models:** multimodal fusion, cross-modal interactions, clinical foundation models
- **Counterfactual Reasoning & Simulation:** counterfactual simulation, physiology-constrained generative models
- **Trustworthy & Enduring Deployment:** uncertainty quantification, sequential monitoring, continual learning
- **Foundations of Machine Learning:** neural scaling laws, time-series sequence models, interpretability

## EDUCATION

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### M.Sc - Ph.D. in Electrical and Computer Engineering

The University of Texas at Austin

August 2023

Austin, TX, USA

Advisor: Joydeep Ghosh    Committee: Aryan Mokhtari, Atlas Wang, Mingyuan Zhou, Nhat Ho

Dissertation: Reliable Predictions for Structured and Corrupted Data

### B.Eng. in Electrical and Electronics Engineering

The University of Edinburgh

July 2017

Edinburgh, UK

Advisor: Jiabin Jia    GPA: 3.92 (graduated with first-class honor)

University of Edinburgh “2+2” Scholarship in 2015 & 2016

## PUBLICATIONS

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**XH** denotes the author; \* denotes equal contribution; ✉ denotes corresponding author.

### Selected Papers

[1] **On the Invariance and Generality of Neural Scaling Laws**

Top ~3% NeurIPS submissions by reviewer rating

**XH**, Z. Liu, S. Saria, and P. Liang

*Preprint*

2026

[2] **FLAME: Adaptive Mixture-of-Experts for Continual Multimodal Multi-Task Learning**

**XH**, S. Chaudhari, T. Ranade, R. Chellappa, and S. Saria

*Preprint*

2026

[3] **Massively Multimodal Foundation Models: A Frameworks for Capturing Interactions with Specialized Mixture-of-Experts**

Top ~10% of accepted papers by reviewer rating

**XH**, H. Chung, J. Ghosh, P. Liang, and S. Saria

*International Conference on Learning Representations (ICLR)*

2026

[4] **QoQ-Med3: Robust Multimodal Clinical Analysis Foundation Model with Reasoning**

W. Dai, J. She, J. Cheong, **XH**, C. Harris, H. Wei, F. Vahedifard, S. Saria, R. Stevens, and P. Liang

*NPJ Digital Medicine*

2026

- [5] **WATCH: Adaptive Monitoring for AI Deployments via Weighted-Conformal Martingales**  
D. Prinster <sup>✉</sup>, **XH** <sup>✉</sup>, A. Liu, and S. Saria  
*International Conference on Machine Learning (ICML)* 2025
- [6] **FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion**  
*Over 149 citations as of July 2026*  
**XH**, H. Nguyen, C. Harris, N. Ho, and S. Saria  
*Neural Information Processing Systems (NeurIPS)* 2024
- [7] **Designing Robust Transformers using Robust Kernel Density Estimation**  
**XH**, T. Ren, T. Nguyen, K. Nguyen, J. Ghosh, and N. Ho  
*Neural Information Processing Systems (NeurIPS)* 2023
- [8] **Architecture Agnostic Federated Learning for Neural Networks**  
D. Makhija, **XH**, N. Ho, and J. Ghosh  
*International Conference on Machine Learning (ICML)* 2022
- [9] **Simultaneously Reconciled Quantile Forecasting of Hierarchically Related Time Series**  
**XH**, S. Dasgupta, and J. Ghosh  
*Artificial Intelligence and Statistics (AISTATS)* 2021
- [10] **Certified Monotonic Neural Networks**  
*Spotlight presentation: 280/9454  $\approx$  2.96%; over 185 citations as of July 2026*  
X. Liu, **XH**, N. Zhang, and Q. Liu  
*Neural Information Processing Systems (NeurIPS)* 2020
- Other Papers**
- [11] **MILM: Large Language Models for Multimodal Irregular Time Series with Informative Sampling**  
H. Chung, S. Li, Y. Wald, **XH**, S. Saria, and J. Ghosh  
*Preprint* 2026
- [12] **EvoMed-Agent: Self-Evolving Reasoners for Multimodal Clinical Sequences via Active Counterfactual Verification**  
**XH**, Y. Wang, C. Chen, H. Chung, S. Li, G. Gudur, W. Dai, P. Liang, and S. Saria  
*In submission* 2026
- [13] **MINT: Multimodal Instruction Tuning with Multimodal Interaction Grouping**  
X. Shan\*, Q. Cao\*, **XH**\*, H. Yu\*, and P. Liang  
*Preprint* 2025
- [14] **Between Linear and Sinusoidal: Rethinking the Time Encoder in Dynamic Graph Learning**  
H. Chung, S. Chaudhari, **XH**, Y. Wald, S. Saria, and J. Ghosh  
*Transactions on Machine Learning Research (TMLR)* 2025
- [15] **SSL-DA: Semi- and Self-Supervised Learning with Dual Attention for Echocardiogram Segmentation**  
L. Lv, **XH**, Z. Sun, Z. Li, X. Wang, T. Jiang, Y. Liu, T. Li, J. Xu, L. You, G. Yao, F. Sun, and J. Xing  
*Journal of Imaging Informatics in Medicine* 2025
- [16] **Quadratic Gating Functions in Mixture of Experts: A Statistical Insight**

- P. Akbarian\*, H. Nguyen\*, **XH**\*, and N. Ho  
*In submission to SIAM Journal on Mathematics of Data Science* 2024
- [17] **On Expert Estimation in Hierarchical Mixture of Experts: Beyond Softmax Gating Functions**  
H. Nguyen\*, **XH**\*, C. Harris, N. Ho, and S. Saria  
*In submission to JMLR* 2024
- [18] **Achieving Fairness Across Local and Global Models in Federated Learning**  
D. Makhija, **XH**, J. Ghosh, and Y. Kim  
*Preprint* 2024
- [19] **Novel Node Category Detection Under Subpopulation Shift**  
H. Chung, S. Chaudhari, Y. Wald, **XH**, and J. Ghosh  
*European Conference on Machine Learning and Data Mining (ECML-PKDD)* 2024
- [20] **Efficient Forecasting of Large Scale Hierarchical Time Series via Multilevel Clustering**  
**XH**, T. Ren, J. Hu, J. Ghosh, and N. Ho  
*International Conference on Time Series and Forecasting (ITISE)* 2023
- [21] **A Novel Control-Variates Approach for Performative Gradient-Based Learners with Missing Data**  
**XH**, J. Hu, and J. Ghosh  
*International Joint Conference on Neural Networks (IJCNN)* 2023
- [22] **Dynamic Combination of Heterogeneous Models for Hierarchical Time Series**  
**XH**, J. Hu, and J. Ghosh  
*International Conference on Data Mining Workshops (ICDMW)* 2022
- [23] **Split Localized Conformal Prediction**  
**XH**, Z. Tang, J. Ghosh, and Q. Liu  
*ICML Workshop on Distribution-Free Uncertainty Quantification* 2021
- [24] **Multi-Pair Text Style Transfer for Unbalanced Data via Task-Adaptive Meta-Learning**  
**XH** and J. Lundin  
*ACL MetaNLP Workshop* 2021
- [25] **Model-Agnostic Explanations using Minimal Forcing Subsets**  
**XH** and J. Ghosh  
*International Joint Conference on Neural Networks (IJCNN)* 2021
- [26] **Transparent Interpretation with Knockout**  
**XH**, Y. Feng, N. Zhang, and Q. Liu  
*ICML Workshop on Human Interpretability in Machine Learning (WHI) [Spotlight]* 2020
- [27] **Sensing Personality to Predict Job Performance**  
S. Lin, S.M. Mattingly, **XH**, P. Audia, et al.  
*CHI Workshop on The Future of Work* 2019
- [28] **Effects of Integrated Intent Recognition and Communication on Human-Robot Collaboration**  
A. Gutierrez, M.L. Chang, **XH**, and K.C. Chang  
*International Conference on Intelligent Robots and Systems (IROS)* 2018

## Patents

### [1] Machine-Learning Based Generation of Text Style Variations for Digital Content Items

J. Lundin, O. Schoppe, **XH**, M. Sollami, B. Lonsdorf, A. Ross, D. Woodward, and S. Rohde

US Patent 2022/0245322 A1

2022

## ACADEMIC EMPLOYMENTS

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**Postdoctoral Fellow**, Johns Hopkins University

Baltimore, MD & Mountain View, CA

Supervisors: Prof. Suchi Saria, Prof. Gregory D. Hager

09/01/2023 – now

- Developing multi-modal Transformers and agents that integrate vital signs, lab results, clinical notes, and chest X-rays, enhancing predictive accuracy/reasoning of multimodal models with an emphasis on clinical applications

**Graduate Research Assistant**, The University of Texas at Austin

06/01/2019 – 08/14/2023

Intelligent Data Exploration and Analysis Lab, Department of Electrical and Computer Engineering

Supervisor: Prof. Joydeep Ghosh

Austin, TX

- Research topics include time-series forecasting, clustering, uncertainty quantification, federated learning, robustness and interpretability of machine learning

Data Science Group, Department of Statistics and Data Sciences

Supervisor: Prof. Nhat Ho

Austin, TX

- Research topics include optimal transport, Bayesian deep learning, model fusion/alignment, mixture of experts, and robustness of Transformers

Statistical Learning and AI Group, Department of Computer Science

Supervisor: Prof. Qiang Liu

Austin, TX

- Research topics include the robustness and interpretability of deep neural networks, conformal prediction, generative models, and predictive modeling of mobile sensor data

## INDUSTRY EMPLOYMENTS

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**Google**

Sunnyvale, CA

Research Scientist Intern (Host: Dr. Xu Gao)

05/31/2022 – 09/02/2022

- Enhanced the workload prediction accuracy for GCP machines by 20% compared to baseline
- Utilized explanation tools (Shapley value) to understand the root causes of machine downtime

**Intuit**

Mountain View, CA

Data Scientist Intern (Mentor: Dr. Jing Hu)

06/15/2021 – 09/03/2021

- Improved cash-flow forecasting accuracy by 10% and deployed models to applications
- Decreased computation time by 45% through the development of a parallelized program

**Salesforce**

San Francisco, CA

Research Intern (Mentor: Dr. Jessica Lundin)

06/01/2020 – 08/21/2020

- Implemented 3 different state-of-the-art language models (LM) for text style transfer
- Leveraged meta-learning to fine-tune pre-trained LM on a small and unbalanced dataset

**CognitiveScale**

Austin, TX

Applied Scientist Intern (Mentor: Dr. Suyog Jain)

05/29/2018 – 08/03/2018

- Reproduced and improved 5 different session-based recommendation algorithms
- Led a team project of 4 interns to research “Ethical AI” and pitched a business idea to company

## GRANTS & PROTOCOLS

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**Grant Drafter and Key Personnel**, Intuit AI Research, **\$300K** in total 2020 – 2023

- Drafted grant proposals and budget planning documents for ~ \$100,000/year from **Intuit AI Research** for three consecutive years. Developed a series of models in time-series forecasting and clustering, deployed in industry-level financial forecasting pipelines. Awarded annually to Dr. Joydeep Ghosh.

**Core Contributor**, Gordon and Betty Moore Foundation Grant, ~ **\$500K** in total 2023 – now

- Developed multimodal clinical foundation models and post-deployment continuous monitoring methods that served as core technical contributions. Awarded annually to Dr. Suchi Saria.

**Protocol Drafter and Submitter**, Johns Hopkins Medicine IRB 2024 – now

- Drafted, submitted, and maintained annual renewals for the following IRB protocols:
  - **IRB00464029**, “Leveraging Multimodal Patient Data for Enhanced Clinical Task Performance”
  - **CR00054024**, “Development and Validation of Robust Models for Adverse Event Risk Prediction”

## COMMUNITY SERVICES

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|                                                                                           |                  |
|-------------------------------------------------------------------------------------------|------------------|
| Reviewer, International Conference on Learning Representations (ICLR)                     | 2020 – now       |
| Reviewer, Neural Information Processing Systems (NeurIPS)                                 | 2020 – now       |
| Reviewer, International Conference on Machine Learning (ICML)                             | 2021 – now       |
| Reviewer, Artificial Intelligence and Statistics (AISTATS)                                | 2021 – now       |
| Reviewer, International Joint Conference on Neural Networks (IJCNN)                       | 2023 – now       |
| Reviewer, Machine Learning for Health (ML4H)                                              | 2023 – now       |
| Reviewer, International Conference on Acoustics, Speech and Signal Processing (ICASSP)    | 2023 – now       |
| Reviewer, Machine Learning for Healthcare (MLHC)                                          | 2024 – now       |
| Reviewer, SIAM Conference on Data Mining (SDM)                                            | 2026 – now       |
| Journal Reviewer, Pattern Recognition                                                     | 2019 – now       |
| Journal Reviewer, Journal of Open Source Software                                         | 2024 – now       |
| Journal Reviewer, IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI) | 2025 – now       |
| Journal Reviewer, Nature Biomedical Engineering                                           | 2025 – now       |
| Journal Reviewer, Medical Image Analysis                                                  | 2026 – now       |
| Mentor, ML4H Author & Reviewer & Career Mentorship Program                                | 2024 – now       |
| Technical Program Committee, WellComp Workshop at UbiComp/ISWC                            | 2026             |
| Reading Group Organizer, Multimodal Learning in Health at JHU                             | 2024 – 2025      |
| Member, UT-Austin ECE Graduate Admission Committee                                        | 2021 – 2023      |
| Volunteer, International Conference on Machine Learning (ICML) Workshops                  | 2020, 2021, 2023 |
| Third-year Class Representative, University of Edinburgh                                  | 2015 – 2016      |

## SUPERVISION

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|                                                   |                                          |
|---------------------------------------------------|------------------------------------------|
| Shravan Chaudhari, CS Ph.D. Student               | JHU, 2023 –                              |
| Carl Harris, Biomedical Engineering Ph.D. Student | JHU, 2023 –                              |
| Chen Chen, CS M.Sc. Student                       | JHU, 2026 –                              |
| Yuxin Wang, Applied Mathematics M.Sc. Student     | JHU, 2026 –                              |
| Jiarui Shao, CS M.Sc. Student                     | JHU, 2026 –                              |
| Tanvi Ranade, CS B.Sc. + M.Sc. Student            | JHU, 2024 – 2026                         |
| Hsing-Huan Chung, ECE Ph.D. Student               | UT-Austin, 2023 – 2026 (now with Uber)   |
| Disha Makhija, ECE Ph.D. Student                  | UT-Austin, 2023 – 2024 (now with Amazon) |

## SELECTED HONOURS AND AWARDS

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|                                                           |             |
|-----------------------------------------------------------|-------------|
| ICML Gold Reviewer Award                                  | 2026        |
| NeurIPS Scholar Award, \$2000                             | 2023 – 2024 |
| Excellent Teaching Assistant Award, \$1000                | 2018        |
| Outstanding Graduates of Beijing City (top 3%)            | 2017        |
| University of Edinburgh 2 + 2 Scholarship, £2500 per year | 2015 – 2016 |
| Comprehensive Scholarship, ¥5000                          | 2014 – 2015 |
| China National Scholarship, ¥8000                         | 2013 – 2014 |

## TEACHING EXPERIENCE

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|----------------------------------------------------------------------|-------------------------------|
| Course Design and Trial Teaching, JHU Teaching Institute Certificate | 2025                          |
| Teaching Assistant, CS 395T, Deep Probabilistic Modeling             | UT-Austin, 2019               |
| Teaching Assistant, ECE 422C, Software Design & Implementation       | UT-Austin, 2018               |
| Lab Instructor, ELEE09024, Digital System Design 3                   | University of Edinburgh, 2016 |

## INVITED TALKS & PRESENTATIONS

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|----------------------------------------------------------------------------------------|-------------------------------------------------------|
| Guiding Mixture-of-Experts in Multimodal Learning: Temporal Dynamics and Robust Fusion | <i>EcoSta</i> , 2025                                  |
| FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion                         | <i>MIT Media Lab</i> , 2025                           |
| FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion                         | <i>Columbia University</i> , 2024                     |
| Reliable Predictions for Structured and Corrupted Data                                 | <i>Fudan University</i> , 2023                        |
| Hierarchical Time Series Forecasting                                                   | <i>IIF Workshop on Forecast Reconciliation</i> , 2023 |
| Reliable Prediction of Treatment Effect Through Time                                   | <i>Intuit AI</i> , 2023                               |
| Time Series Reconciliation with Dynamically Changing Structures                        | <i>Google Cloud AI</i> , 2022                         |
| Mixture-of-Experts for Probabilistic Forecasts of Aggregated Time Series               | <i>Intuit AI</i> , 2021                               |
| Certified Monotonic Neural Networks                                                    | <i>Salesforce Research</i> , 2020                     |
| Transparent Interpretation with Knockout                                               | <i>Intuit AI</i> , 2019                               |

## TECHNICAL SKILLS

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**Programming:** Python (expert); R, Matlab (proficient); Java, Verilog, C/C++ (knowledge of)  
**Frameworks:** PyTorch, TensorFlow, Scikit-Learn, Wandb, HuggingFace, TIMM, NLTK  
**Language:** English (fluent), Mandarin (native), French (beginner)  
**Others:**  $\LaTeX$ , Shell Script, ROS, SQL

## REFERENCES

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|                                                                      |                              |
|----------------------------------------------------------------------|------------------------------|
| Suchi Saria, John C. Malone Associate Professor (Postdoc Advisor)    | ssaria@cs.jhu.edu            |
| Department of Computer Science, Johns Hopkins University             | Baltimore, MD & New York, NY |
| Joydeep Ghosh, Schlumberger Centennial Chair Professor (PhD Advisor) | jghosh@utexas.edu            |
| Department of Electrical and Computer Engineering, UT-Austin         | Austin, TX                   |
| Paul Pu Liang, Assistant Professor                                   | ppli@mit.edu                 |
| Media Lab and EECS, MIT                                              | Cambridge, MA                |